Exp:1 Study of various Network Commands used in Linux and windows
28/7/26

Aim:

Study of various Network Commands used in Linux and windows

Linux Networking Commands:

① ifConfig:

To find troubleshooting networks and configuring

output:

enp0s3: flags=4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::a00:27ff:fe17:ea56 Prefixlen 64 Scope id 0x20<link>

lo: flags=73<UP, LOOPBACK, RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.000.0
inet 6::1 Prefixlen 128 scope id 0x10<host>
loop txqueuelen 1000 (Local Loopback)

② mtr:

It provides response time and percentage the statistics of the website.

a) mtr google.com: shows you the statistics, including each hop with time and loss %.:

output:

Google.Com 2026-07-26T22:00:45+0530

Keys: Help displaymode Restart Statistics order of fields Ping

Host	Loss%	Snt	Last	Avg	Best	Wrst
1.-gateway	0.0%	592	0.6	2.2	0.4	25.9
2. fnmaaa-av-in-f14.1e100.net	0.0%	658	93.8	55.2	24%	937.6

b) mtr -g google.com:

It is similar to guide

output:

mtr [options] hostname

-F, --filename File	read hostname(s) from a file
-4	use IPv4 only
-6	use IPv6 only
-u, --udp	use UDP instead of ICMP echo
-T, --tcp	use TCP instead of ICMP echo

c) mtr -b google.com:

It is similar to mtr google.com

output:

google 2026-07-26T22:13:59

Keys: Help Display mode Restart statistics order of fields quit

Host	Packets		Pings			
	Loss%	Snt	Last	Avg	Best	Wors
1.- gateway (10.0.2.2)	0.0%	413	0.8	2.1	0.3	22.1
2. bKK02501-in-f14.1e100.net	0.0%	505	25.4	6.7.1	21.1	

d) mtr -C 10 google.com:

It is similar to mtr -b google.com

output:

Same output as mtr -b google.com

③ tcpdump:

Capturing and displaying Packets.

a) dnf install -y tcpdump:

to start before Capturing packets

b) tcpdump -D:

List all Connections

output:

1. enpos3 [UP, Running, Connected]
2. any (Pseudo-device that Captures on all interfaces) [UP, Running]
3. lo [up, Running, Loopback]
4. bluetooth-monitor (Bluetooth Linux Monitor) [wireless]

c) tcpdump -i enpos3:

It will Capture traffic on enpos3

d) tcpdump -i enpos3 -c 10:

Capture traffic to and from one host

e) tcpdump -i ~~eth0~~ enpos3 src host 8.8.8.8:

Same to tcpdump -i enpos3

f) tcpdump -i enpos3 -c 10 host www.google.com and port 443:

To Capture only HTTPS traffic

④ Ping: verifies IP-level Connectivity to another TCP/IP Computer by Sending Internet Control message Protocol (ICMP) Echo Request messages.

output

64 bytes from lemaaa-az-in-f14.1e100.net (142.251.221.110): icmp_seq=1 ttl=111 time=31.7ms

" " " "

output:

windows IP Configuration

Ethernet adapter Ethernet 2:

Connection-Specific DNS Suffix . :

Link-local IPv6 Address . . . : fe80::c8e6:8ba1:700:5%bd%2

IPv4 Address . . . . . : 192.168.56.1

Subnet Mask . . . . . : 255.255.255.0

④ nbtstat -a:

Displays protocol statistics and current TCP/IP connections using NBT manual

⑤ netstat:

network statistics active TCP/IP connections

output:

active Connections

Proto	local address	foreign address	state
TCP	10.202.7.87.49285	10.202.7.160:domain	Time-wait
TCP	10.202.7.87.49783	10.202.7.160:domain	Time_wait
Tcp	10.202.7.87.50583	10.202.7.160:domain	Time-wait
Tcp	10.202.7.87.50595	10.202.7.160:domain	Time-wait

⑥ nslookup:

name sever lookup is a tool used to perform DNS lookups in linux. In two modes interactive and non-interactive

a) nslookup www.google.com

output:

Server: 10.202.7.160

address: 10.202.7.160#53

Non-authoritative:

Name: google.com

address: 142.251.157.119

Name: www.google.com

address: 142.251.151.119

Name: www.google.com

address: 142.251.154.119

⑦ pathping:

Combination of the Ping and Tracert commands manual

⑧ Route:

shows/manipulate the IP routing table

output:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref
default	- gateway	0.0.0.0	UG	100	0
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0

Result:

All the Commands were executed successfully.